

CUBIC SPLINE INTERPOLATION IN NORMED SPACES

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Objectives and Brief Overview of Research

- Discuss the aim of the thesis (draw ideas from abstract)
- Mention error bounds
- Mention sort analysis and results

What are Cubic Splines and Why are they Important?

- Discuss applications of splines (use Baltzar or Andrew Hsu)
- Mention how splines are useful in computer science in data prediction as mentioned in thesis

General Theory of Cubic Splines

- Defn of both Kreyszig and Mhaskar-Pai splines.
- Define norm before the above.
- Briefly explain how the algorithms differ.

Cubic Spline Algorithm Graphs

- Kreyszig

- Mhaskar-Pai

Next Steps: Applying the Theory to Writing Code

- Explain each algorithm's interpolation conditions.
- Idea: Show a code snippet and in-presentation relate the interpolation conditions to the code.

Data Collection: Observations and Conjectures

- Summarize data collection procedure
- Discuss what types of algorithms were used for built-in Mathematica functions (if time allows)

Concluding Remarks and Future Work

- Summarize and conclude the analysis and results from it.
- Mention future work both immediate and far future with emphasis on immediate.

— THANK YOU FOR LISTENING !